INTRODUCTION TO COMPUTER NETWORKING

NETWORK

A network, is a group of two or more devices that can communicate. A network is contain of a number of different computer systems connected by physical or wireless connections.

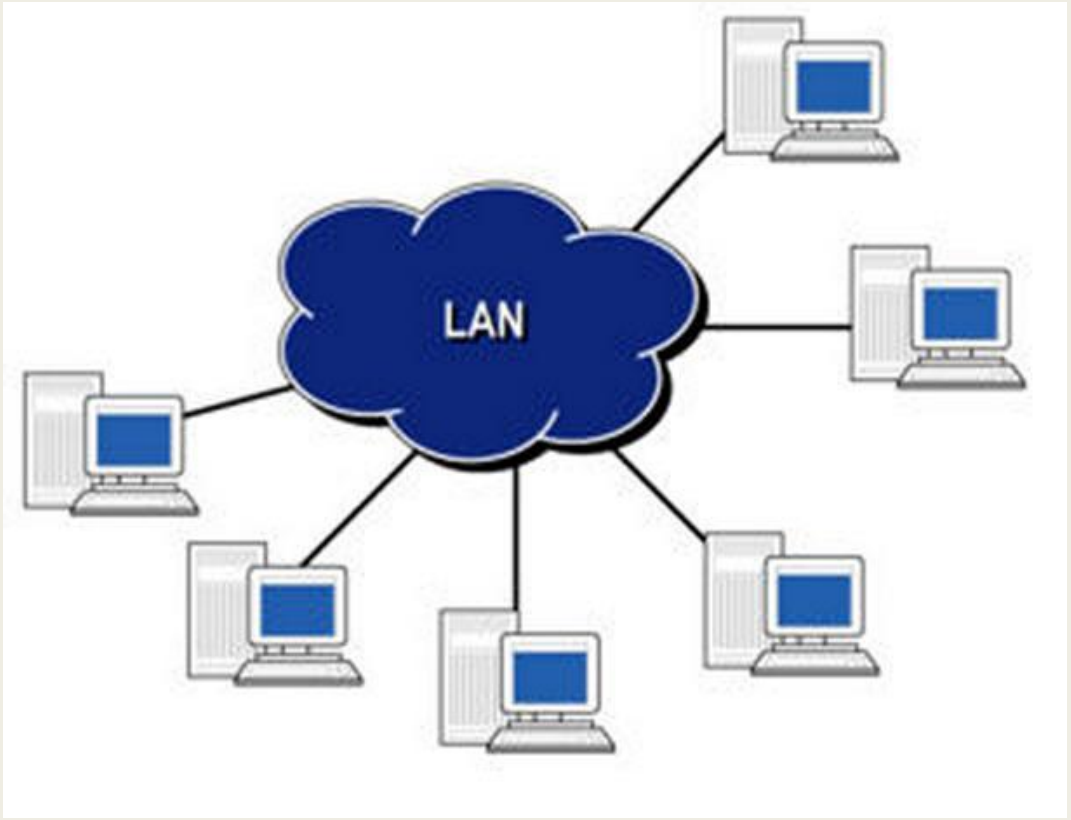


TYPES OF NETWORK

- ❑ **Local Area Network (LAN)**
- ❑ **Wireless Local Area Network (WLAN)**
- ❑ **Metropolitan Area Network (MAN)**
- ❑ **Wide Area Network (WAN)**

Local Area Network (LAN)

A LAN is a network of connected devices that exist within a specific location. LANs may be found in homes, offices, educational institution, or other areas.



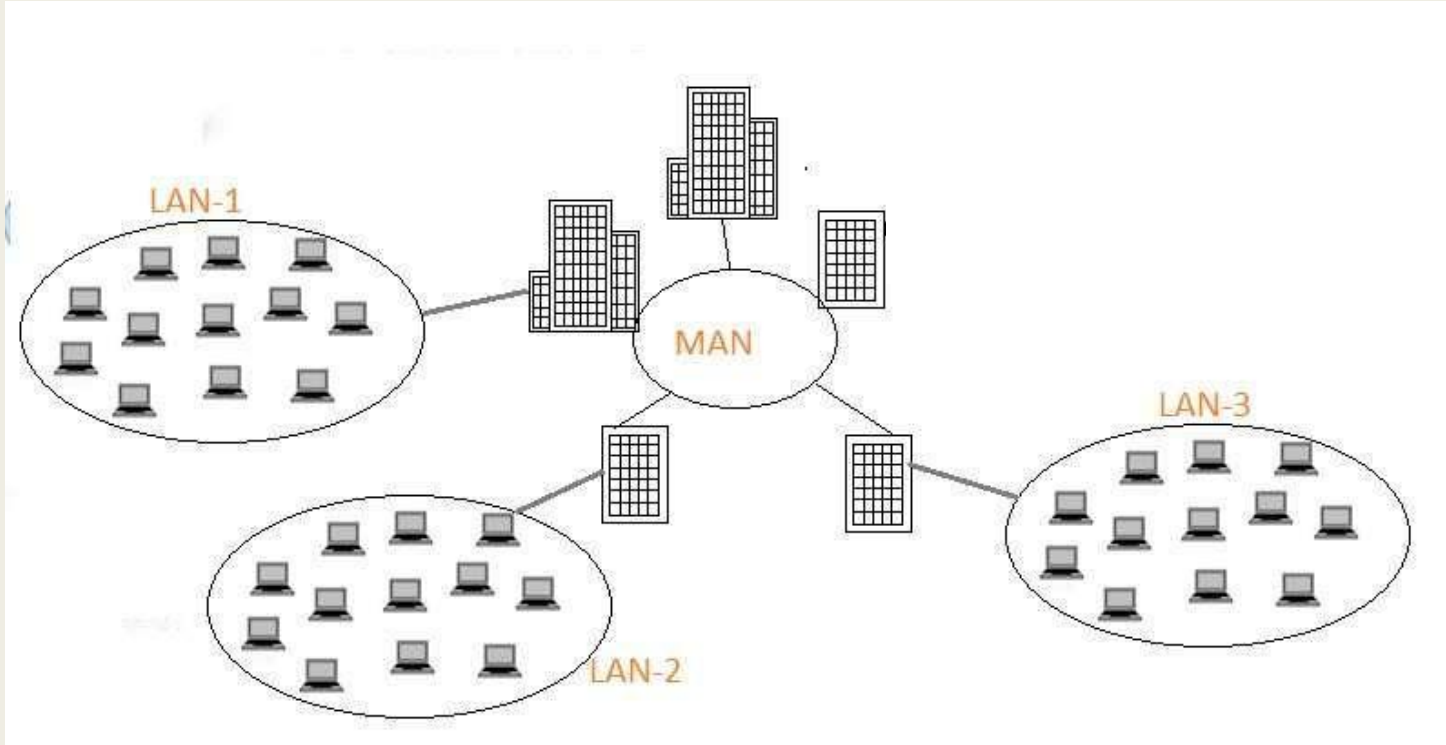
Wireless Local Area Network (WLAN)

A wireless local area network (WLAN) is a wireless distribution method for two or more devices that use high-frequency radio waves and often include an access point to the Internet.



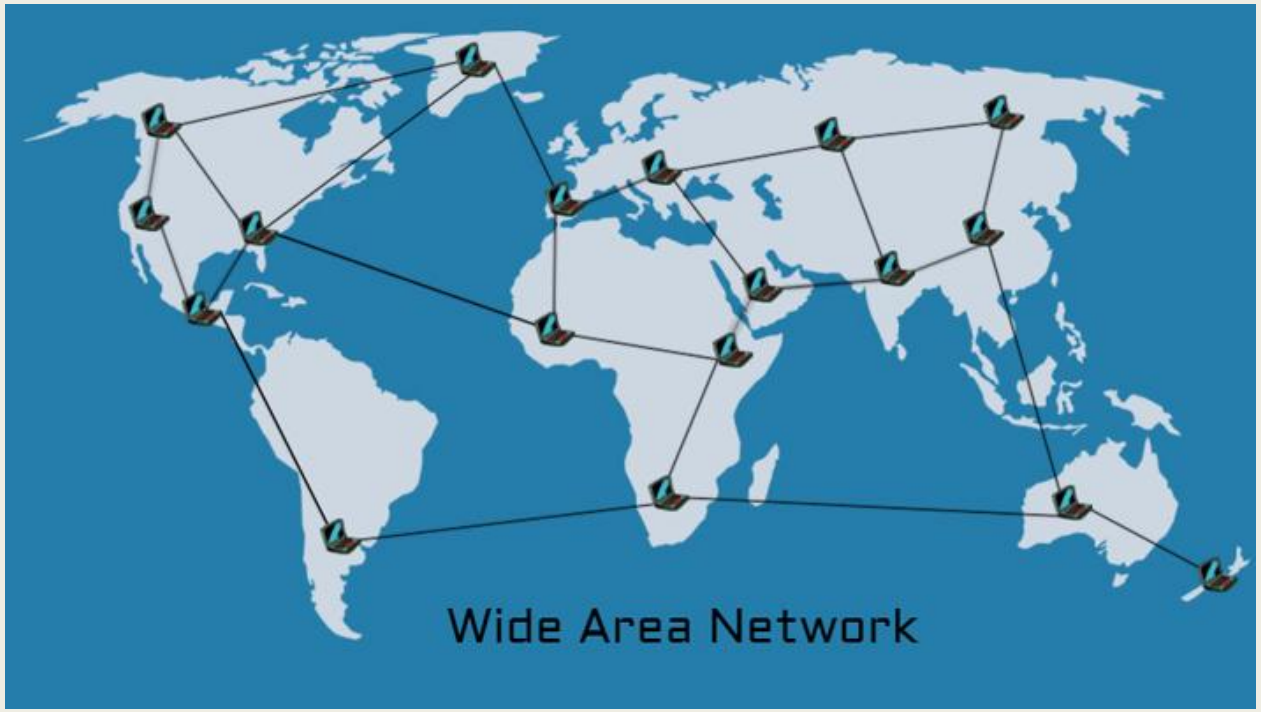
Metropolitan Area Network (MAN)

A metropolitan area network (MAN) is similar to a local area network (LAN) but spans an entire city or campus. MANs are formed by connecting multiple LANs.



Wide Area Network (WAN)

A wide area network (WAN) is a network that exists over a large-scale area. A WAN connects different smaller networks, including local area networks (LANs) and metro area networks (MANs).



NETWORKING

Computer Networking is the practice of transporting and exchanging data between nodes over a shared medium in an information system. Networking is not only the construction and use of a network, but also the management, maintenance and operation of the network.



Internet Protocol (IP)

An Internet Protocol address (IP address) is a numerical label like (192.168.0.1) assigned to each device connected to a computer network that uses the Internet Protocol for communication.

Media Access Control (MAC) Address

A MAC address is given to a network adapter when it is manufactured. It is hard-coded onto your computer's network interface card (NIC) and is unique to it.

Example: 3A-34-52-C4-69-B8

Network Switch

A network switch determines where to send each incoming message frame by looking at the physical device address (also known as the Media Access Control address or MAC address).



Wireless Router

A wireless router is an electronic device that works as a router meaning it sends data from the internet cable to a device and as a wireless access point so this data can be shared through radio signals instead of another cable.





Network Cable

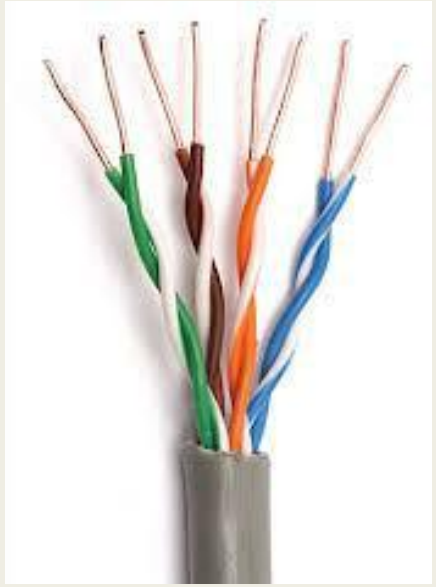
Cable is the medium through which information usually moves from one network device to another.

Basic Type of Cables

- **Unshielded Twisted Pair (UTP) Cable**
- **Shielded Twisted Pair (STP) Cable**
- **Fiber Optic Cable**
- **Coaxial Cable**

Types of Cables

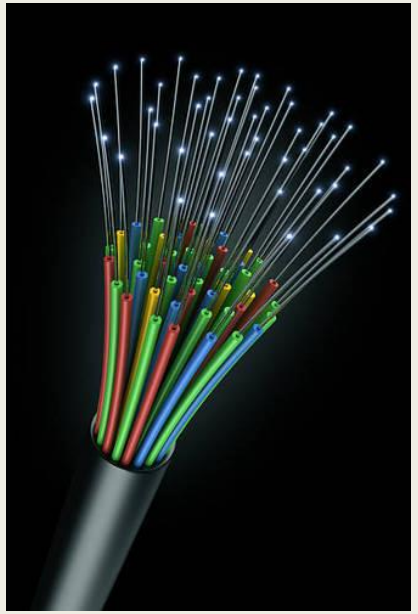
Unshielded Twisted Pair



Shielded Twisted Pair



Fiber Optic Cable



Coaxial Cable



Types of Connectors

- **RJ-45 for Shielded and Unshielded Twisted Pair cables**
- **Subscriber or Standard Connector (SC) for Fiber Optic cables**
- **F-Connector or F-Type Connector for Coaxial cables**



RJ-45

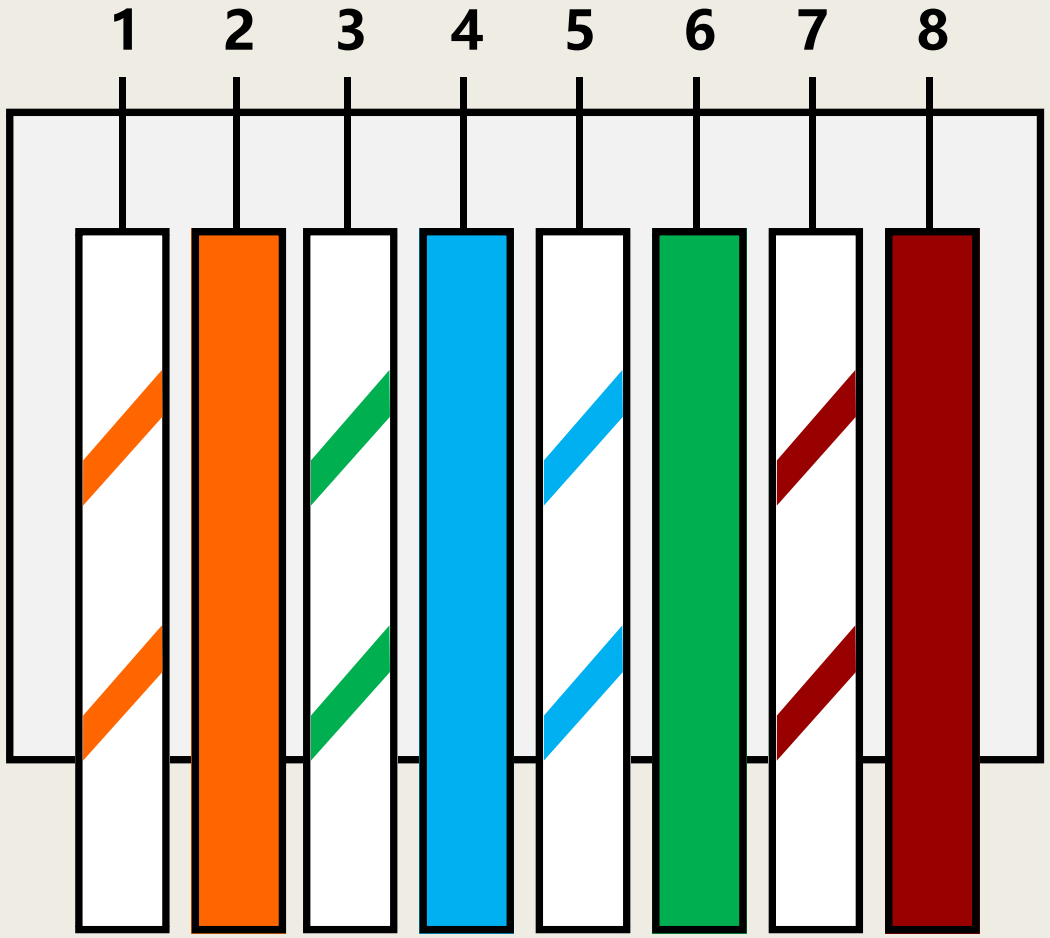


SC



F-Type

Straight Pair for Twisted Pair Cable

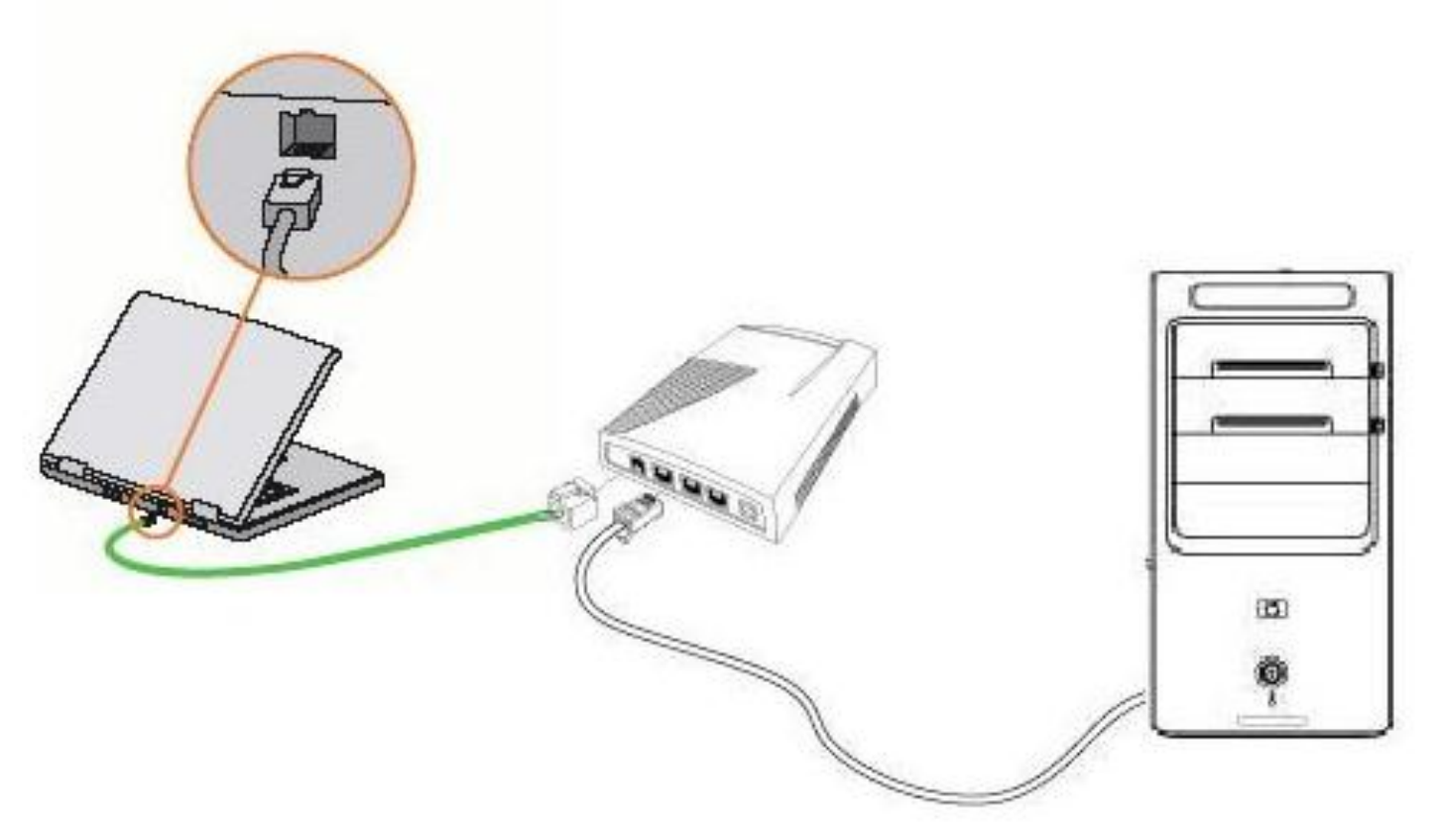




HOW TO SHARE FILES AND FOLDERS

Connecting the Network

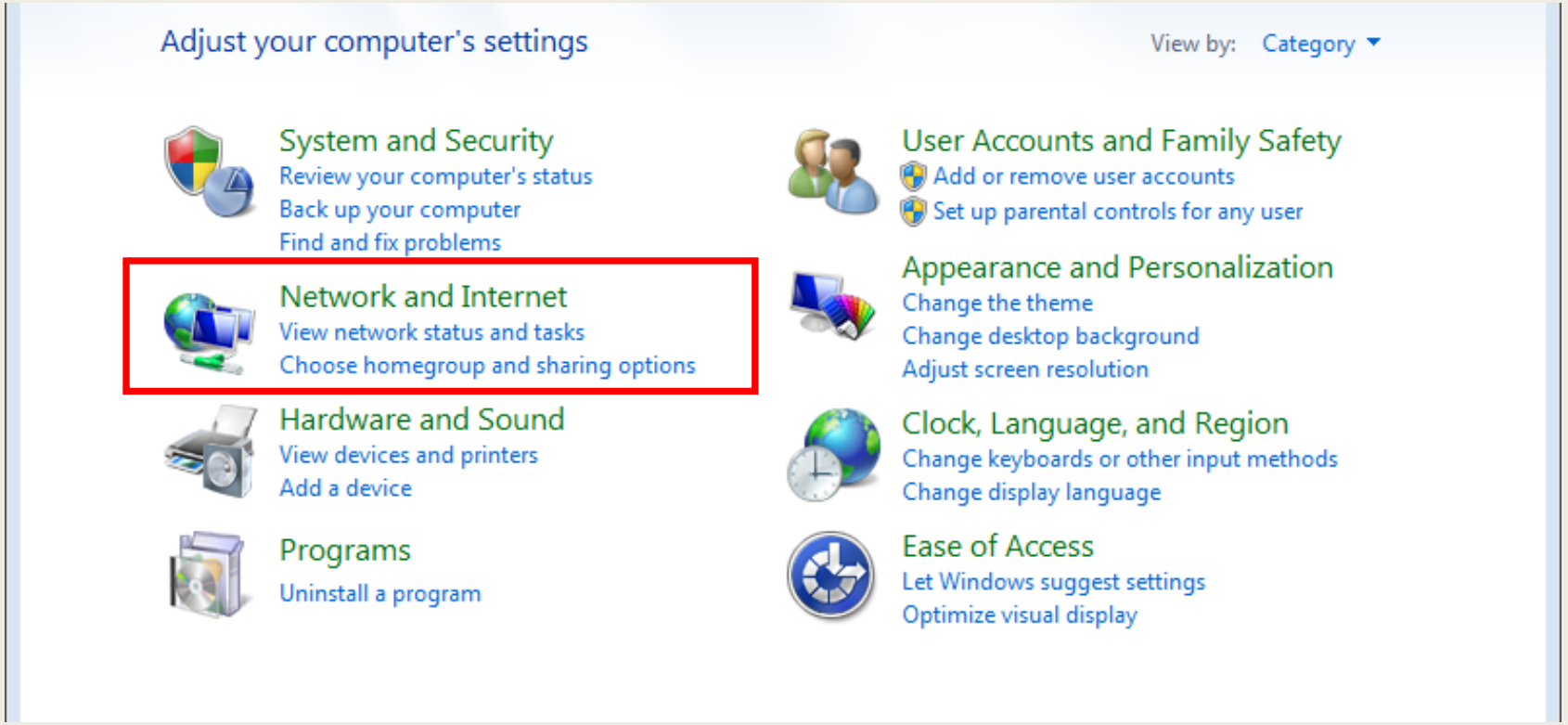
1- Connect Both Devices on a same network.



Turning on Network discovery

1-Click START Button, and then click Control Panel.

2-Under Network and Internet, click Choose Homegroup and Sharing options.



Turning on Network discovery

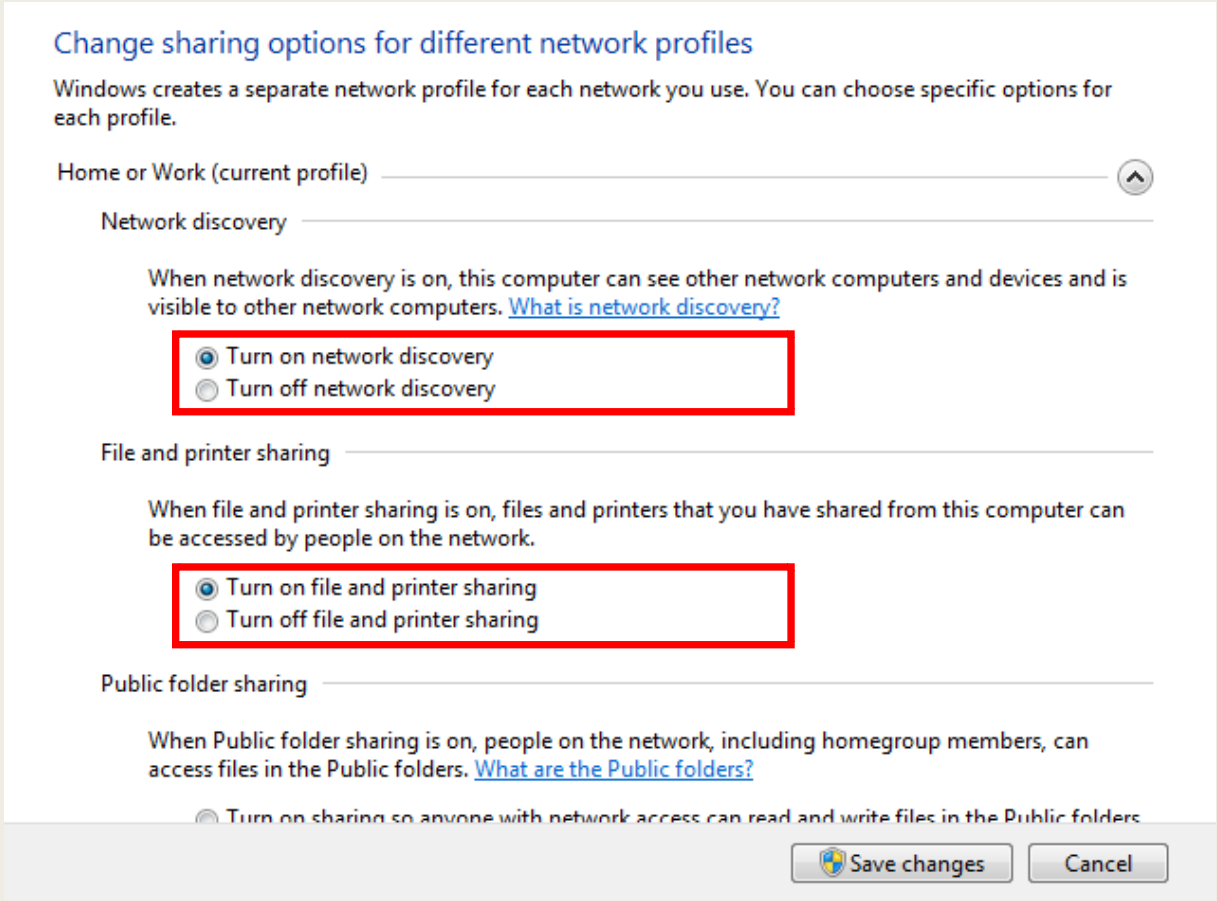
3-In the Homegroup setting windows, click Change advanced sharing settings.



Turning on Network discovery

4-Turn on Network discovery and File and Printer Sharing.

5-Click Save changes.

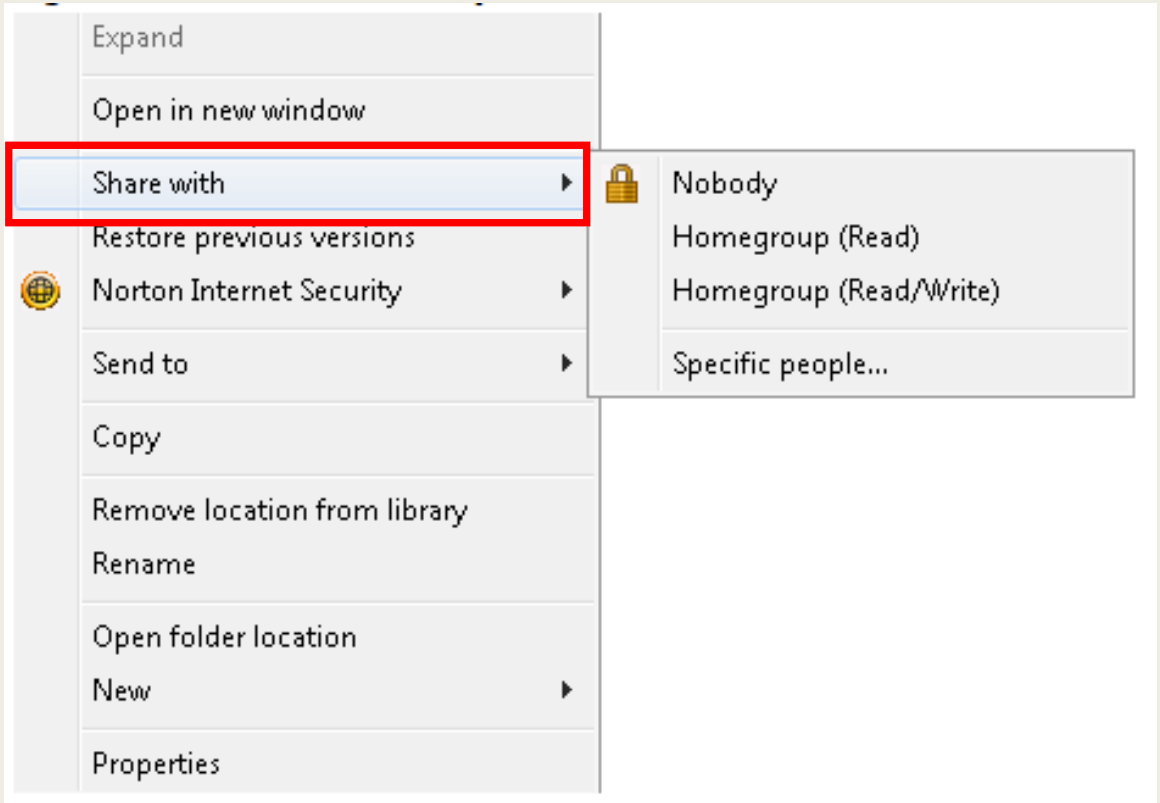


Sharing folders in a Windows 7 network

1- Open Windows Explorer.

2- Browse to the folder you want to share.

3- Right-Click the folder, Select Share with, and then click Homegroup (Read), Homegroup (Read/Write), or Specific people.



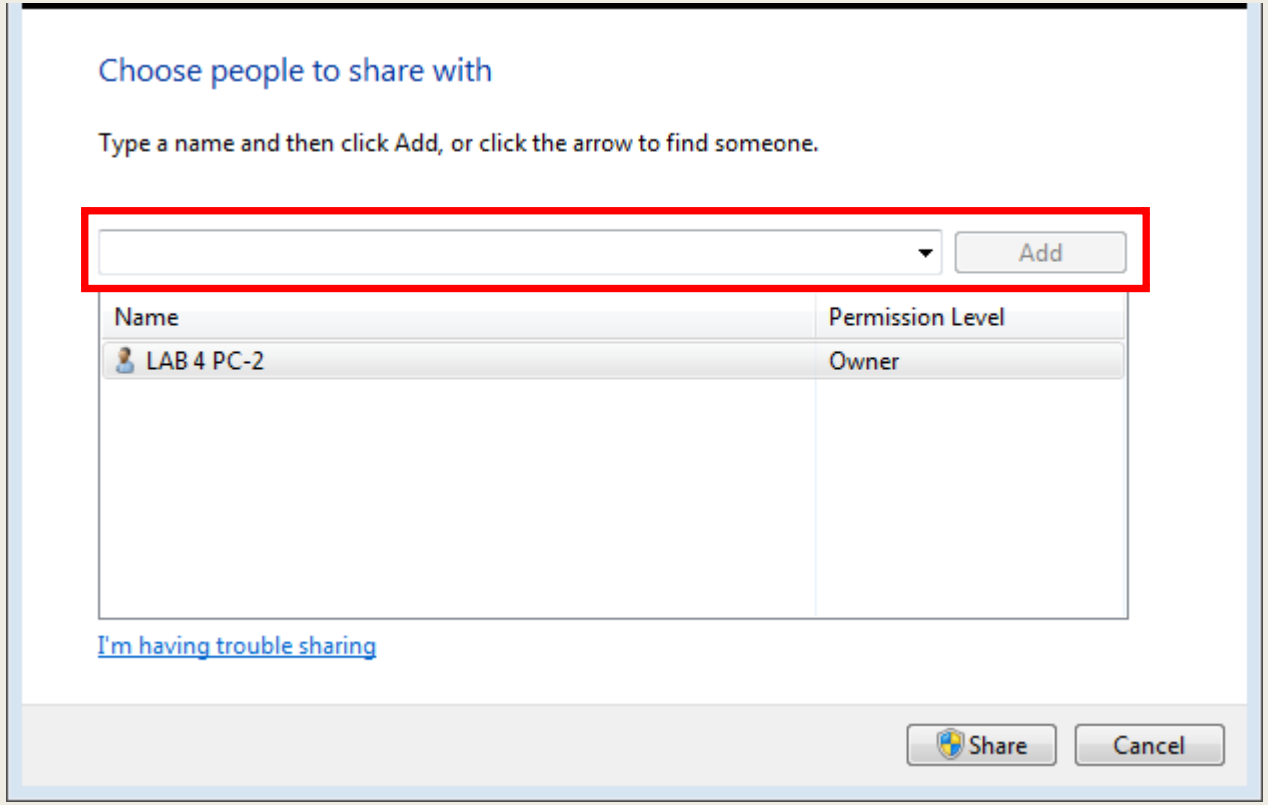
Sharing folders in a Windows 7 network

4- If you chose specific people, the File Sharing window displays.

5- Click the down arrow and select the account you want to share with, and then click Add.

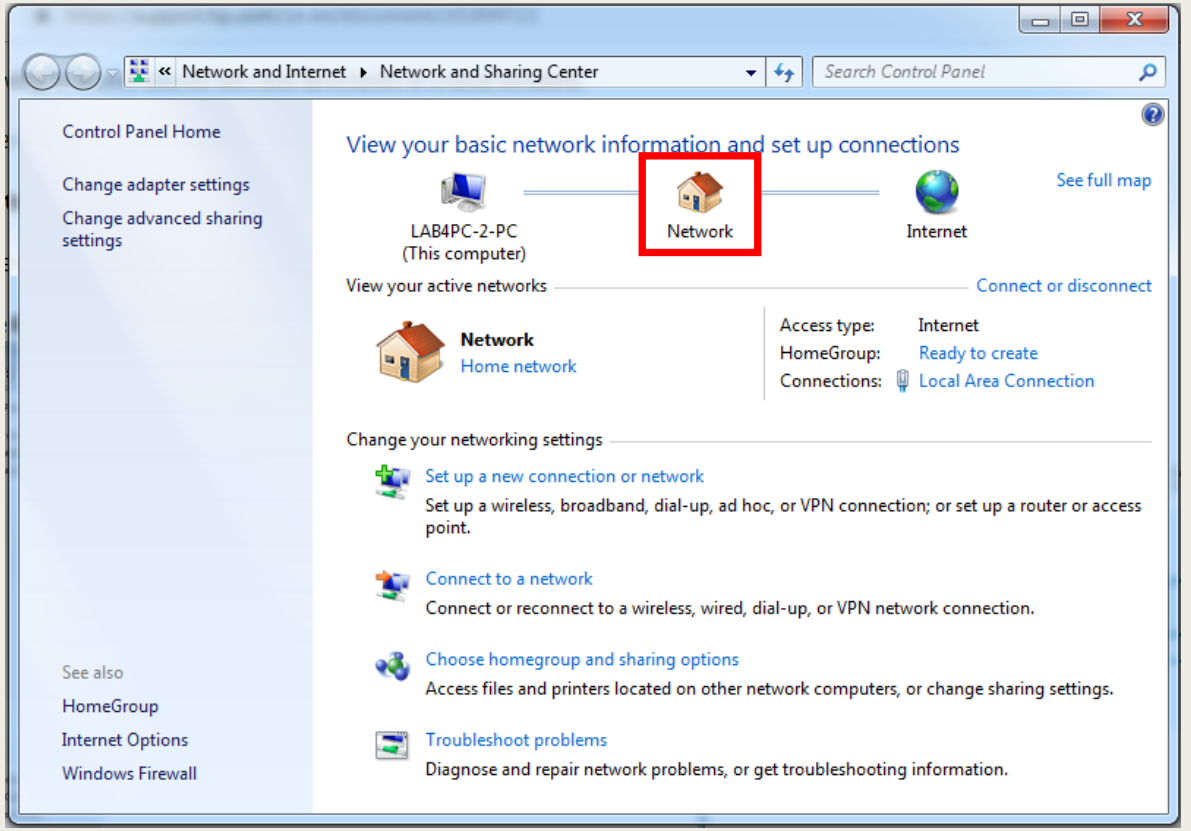
6- Click an arrow under Permission Level to set the permission level for each account or group.

7-Click Share.



Accessing shared files and directories

1- Ensure Network discovery and File Sharing is turned ON.
2- Click Start Button, Click Control Panel, Click Network and Internet, and then click Network and Sharing Center.
3- Double-click Network.



Accessing shared files and directories

4- The Network window opens and displays computer with shared folder that are detected on local networks.

5- Double-click the computer you want to access.

